

We construct the treasury curve from the treasury instruments to provide the basic interest rate components:

- Discount factor
- Forward rate
- Zero rate.

These components are the main building blocks for pricing and sensitivities calculation of fixed income instruments.

We describe the methodology of the treasury curve construction. Two curve building methodologies are used to construct treasury discount curve, one is the bootstrap method, and the other one is optimization method. The bootstrap method iteratively solves the discount factors at pillar dates (instruments maturity dates) from short term instruments to long term instruments to price back the curve instruments one by one. The optimization method targets to solve the array of discount factors at pillar dates such that the sum of absolute value of pricing errors stays below a tolerance range (e.g. less than $10e-12$).

To construct a treasury curve, interpolation method is an important model assumption. Different interpolations methods like LOGLINEAR, LOGPARABOLIC, LOGCUBIC and mixture interpolation of LOGLINEAR and LOGCUBIC can be used to derive the discount factors between pillar dates. And a constant instantaneous forward rate extrapolation is adopted to extend the treasury curve beyond the last pillar date.

Notations

T_0 : Trade date (same as curve base date where $D(T_0) = 1$)

T_{issue} : The issuance date of treasury bills/notes/bonds.

$T_{settlement}$:	Settlement date
T_M :	Trade notional payment date
$T_{last\ coupon}$:	The last coupon accrual end date.
$\Delta_{day\ count}(t_1, t_2)$:	The year fraction between date t_1 and date t_2 under specified day count convention.
$\{T_i\}_{i=1 \dots M}$:	The coupon payment schedule after $T_{settlement}$
$\{T_i^S\}_{i=1 \dots M}$:	The coupon accrual start dates since last coupon.
$\{T_i^E\}_{i=1 \dots M}$:	The coupon accrual end dates since last coupon.
MV_{clean} :	The market quote of note/bond clean price
MV_{bill} :	The market settlement value for treasury bill.
$D(t)$:	The discount factors at date t
y_i :	The treasury bill yield
c :	The coupon rate for treasury notes/bonds
N :	The treasury bill/note/bond notional.
$SV_{clean}(t)$:	The calculated note/bond clean price (settlement value) settled at date t .
$PV_{dirty}(t)$:	The calculated note/bond dirty price settled at date t .
$SV_{bill}(t)$:	The calculated bill price (settlement value) settled at date t .

Instruments

We use US Treasury curve as an example to illustrate the curve instruments. The instruments used for the US treasury curve construction are treasury bills and treasury notes/bonds. By taking the market quotes of those two types of instruments, the treasury curve can be bootstrapped accordingly.

Treasury Bill The treasury bill (T-Bill) is a short-term debt obligation backed by the U.S. government with a maturity of less than one year. Like zero-coupon bonds, they do not pay interest prior to maturity; instead they are sold at a discount of the par value. The standard market quote of Treasury bill is based on the Treasury bill yield, which is defined below.

$$MV_{bill} = 100 \cdot (1 - y \cdot \Delta_{ACT/360}(T_{settlement}, T_{maturity})) \quad (1)$$

Given the fact that treasury bills can be traded before the issuance, the settlement date $T_{settlement}$ is $\max(T_0 + 1B, T_{issue})$, here $T_0 + 1B$ means the next business (under US bond holiday calendar) after trade date.

For treasury curve construction, the following formula is adopted to price the zero coupon bond.

$$PV_{bill} = 100 \cdot \frac{D(T_M)}{D(T_{settlement})} \quad (2)$$

Treasury Notes/Bonds US treasury notes (T-Notes) mature in two to ten years. The treasury bonds (T-Bonds) have the longest maturity, from twenty years to thirty years, and are commonly issued with maturity of thirty years. Both treasury notes and bonds have a coupon payment every six months. For the treasury notes and bonds, the dirty price is defined in the following:

$$\begin{aligned} PV_{dirty}(T_{settlement}) &= \frac{1}{D(T_{settlement})} (N \cdot D(T_M) \\ &+ \sum_{i=1}^M N \cdot c \cdot \Delta_{ACT/ACT.ICMA}(T_i^S, T_i^e) \cdot D(T_i)) \end{aligned} \quad (3)$$

Under the ACT/ACT.ICMA day count convention, the year fraction between accrual start and end dates is defined simply as $\Delta_{ACT/ACT.ICMA}(T_i^S, T_i^e) = \frac{1}{\text{coupon Frequency}} = 0.5$. The coupon payment date T_i is derived by adjusting T_i^e to business day under “following” Business Day Convention, i.e. if T_i^e is a business day, $T_i = T_i^e$, otherwise, T_i is the next business day after T_i^e .

The treasury notes and bonds are commonly traded based on the clean price at settlement date, which is the dirty price excluding the accrual amount from last coupon payment to settlement date.

$$\begin{aligned} PV_{clean}(T_{settlement}) &= PV_{dirty}(T_{settlement}) - N \cdot c \\ &\cdot \Delta_{ACT/ACT.ICMA}(T_{last}, T_{settlement}) \end{aligned} \quad (4)$$

Similar as the treasury bill, the settlement date $T_{settlement}$ is $\max(T_0 + 1B, T_{issue})$ for treasury notes/bonds.

Market Quote Details

For the treasury instruments market quotes, we use the Bloomberg data. The interest rates used and conventions are listed in the Table 1.

Table 1: Data Source for Treasury Curve

This table lists the treasury bills and notes/bonds data sources for calibrating the US treasury curve. Treasury bills are used in curve construction are from 1 month up to 1 year. Treasury notes are used from 2 years up to 10 years. We use only one liquid treasury bond which matured in 30 years.

TREASURY MARKET DATA						
BLOOMBERG TICKER	TYPE	QUOTE TYPE	ACCRUAL DAYCOUNT	SETTLEMENT DAYS	PAYMENT BDC	PAYMENT CALENDAR
CBM Govt	BILL	YIELD	ACT/360	1	FOLLOWING	NEWYORK
CB3 Govt	BILL	YIELD	ACT/360	1	FOLLOWING	NEWYORK
CB6 Govt	BILL	YIELD	ACT/360	1	FOLLOWING	NEWYORK
CB12 Govt	BILL	YIELD	ACT/360	1	FOLLOWING	NEWYORK
CT2 Govt	NOTE	CLEAN PRICE	ACT/ACT.ICMA	1	FOLLOWING	NEWYORK
CT3 Govt	NOTE	CLEAN PRICE	ACT/ACT.ICMA	1	FOLLOWING	NEWYORK
CT5 Govt	NOTE	CLEAN PRICE	ACT/ACT.ICMA	1	FOLLOWING	NEWYORK
CT7 Govt	NOTE	CLEAN PRICE	ACT/ACT.ICMA	1	FOLLOWING	NEWYORK
CT10 Govt	NOTE	CLEAN PRICE	ACT/ACT.ICMA	1	FOLLOWING	NEWYORK
CT30 Govt	BOND	CLEAN PRICE	ACT/ACT.ICMA	1	FOLLOWING	NEWYORK

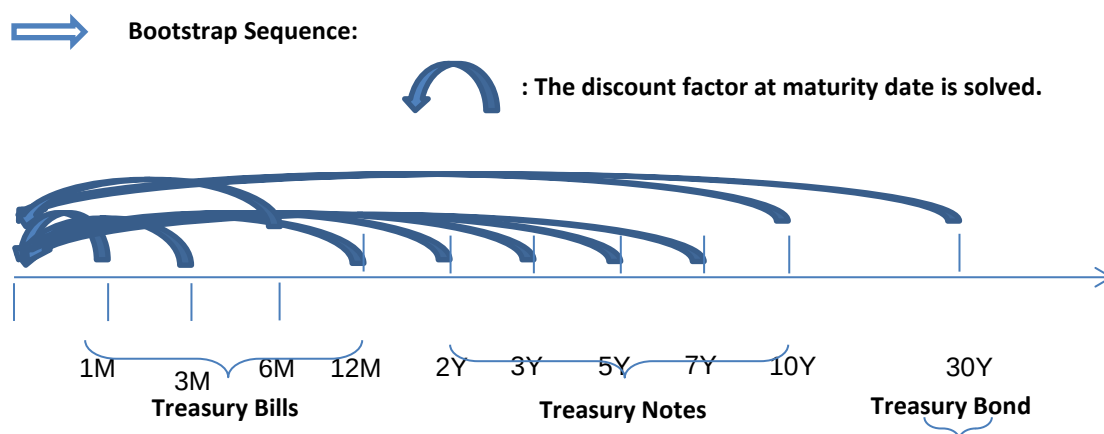
Bootstrap Methodology

The treasury curve bootstrap involves the process of calculating the discount factors from the treasury bills and notes/bonds iteratively from the short end to the long end. For each time period, the previous discount factors are used to interpolate the discount factors in the period, as illustrated in the **Error! Reference source not found.** Treasury Curve Bootstrap Methodology. The discount factor is solved by the Brent's method. The Brent's method is a popular root-finding algorithm, which combines the Bisection method and inverse quadratic interpolation method. Once the accuracy, initial guess and interval are given, the algorithm will try to find the root by using the potentially fast-converging inverse quadratic interpolation if possible, but it falls back to the more robust bisection method if necessary. In this case, the root to be solved is the instantaneous

forward rate between two adjacent instruments and the objective is to price all instruments value $PV(T_{settlement})$ to market settlement value $MV(T_{settlement})$.

Treasury Curve Bootstrap Methodology.

This figure describes the iterative procedure of curve bootstrapping from the short end to the long end. The discount factors at pillar nodes will be updated from the left to the right one by one for each instrument.



The bootstrap methodology is a robust and accurate method for curve construction. However, due to the fact that construction process is moving from short end to long end, once the discount factor is solved, it moves to the next one (*i.e.* the previous values will not be changed at later stage). Given this fact, when a global interpolation method is applied (e.g. cubic spline and monotonic convex), the discount factor values at later pillar dates will change the way it interpolates between earlier pillar dates, which will possibly cause the bootstrap failure to meet the required accuracy.

Hence we introduce the following curve building method using global optimization.

Optimization Curve Building Methodology

The idea of the curve construction is to search for an array of discount factors values D , which will price (discount) the market instruments at their

quoted values under specific interpolation assumption. In other words, we are dealing with the below minimization problem:

$$\min_D \left(\sum_{i=1}^M |MV_i - SV_i(D)| \right) \quad (5)$$

If we found a solution which minimizes the above target function to the tolerance requirement (e.g. 10e-12), the target array of discount factors will be the treasury curve we want.

To solve the above problem, we use the Levenberg-Marquardt Algorithm (LMA). The LMA is a method combines both the Gauss-Newton algorithm and gradient descent, and is able to provide a robust solution to the optimization problem. The key steps of LMA are the following:

1. Compute the Jacobian matrix from the target function.
2. Derive the improvement factor by using LU matrix solver.
3. Continue the step 1-2 until the required tolerance level is met.

User can refer [1] for details of LMA implementation.

Due to the fact that LMA does not specify an order of solving discount factors, it can handle any interpolation method (if a solution exists) during the iteration process.

1.1.1 Interpolation Methodology

Four interpolation methods are provided for the treasury curve construction:

1. LogLinear: Apply linear interpolation to the log value of discount factors.
2. LogParabolic: Apply the parabolic interpolation to the log value of discount factors.
3. LogCubic: Apply the cubic spline interpolation to the log value of discount factors.
4. LogLinearCubic: Apply linear interpolation to the pillar dates less and equal than one year (treasury bill nodes), and apply cubic spline interpolation to the pillar dates more than one year.

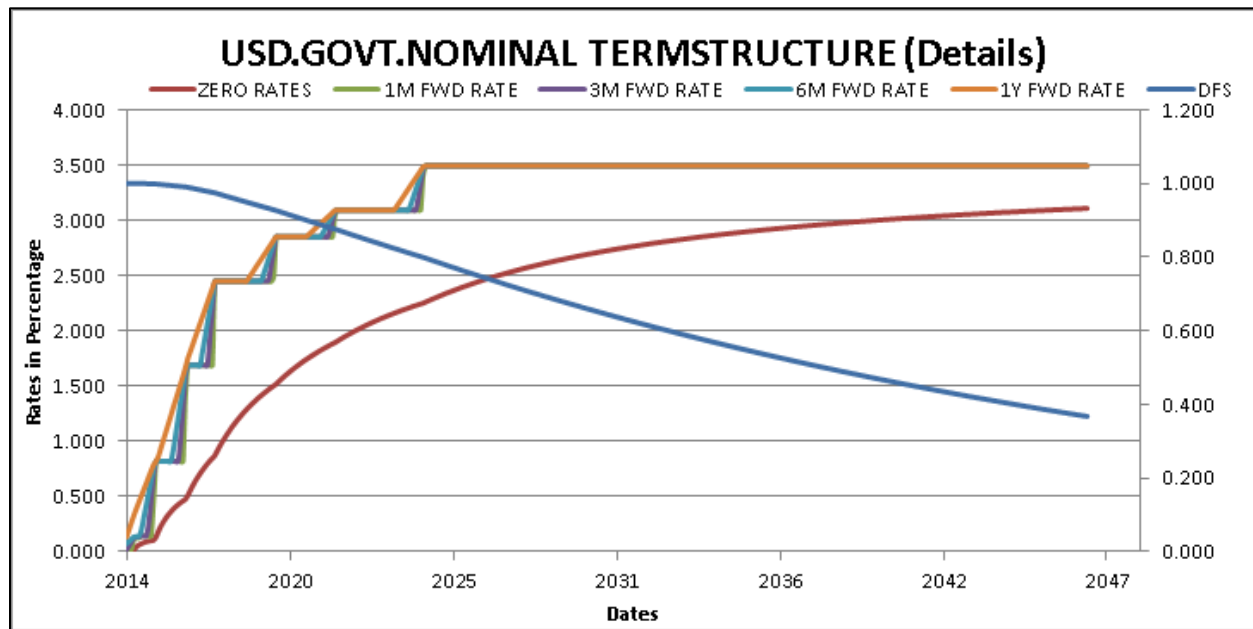
For the above four interpolation methods, LogLinear is implemented with the bootstrap method; the rest 3 methods are implemented with optimization method.

Treasury Curve Calibration

TREASURY MARKET DATA							
USE?	INSTRUMENTS	TYPE	DATED DATE	MATURITY DATE	COUPON	QUOTE TYPE	QUOTE VALUE
Y	CBM Govt	BILL	2014-06-26	2014-12-26	-	Y	0.0150
Y	CB3 Govt	BILL	2014-08-28	2015-02-26	-	Y	0.0150
Y	CB6 Govt	BILL	2014-05-29	2015-05-28	-	Y	0.0700
Y	CB12 Govt	BILL	2014-11-13	2015-11-12	-	Y	0.1100
Y	CT2 Govt	NOTE	2014-11-30	2016-11-30	0.5000	P	100-01 1/4
Y	CT3 Govt	NOTE	2014-11-15	2017-11-15	0.8750	P	100-00 +
Y	CT5 Govt	NOTE	2014-11-30	2019-11-30	1.5000	P	99-31 1/4
Y	CT7 Govt	NOTE	2014-11-30	2021-11-30	1.8750	P	100-01
Y	CT10 Govt	NOTE	2014-11-15	2024-11-15	2.2500	P	100-12 +
Y	CT30 Govt	BOND	2014-11-15	2044-11-15	3.0000	P	101-08 +

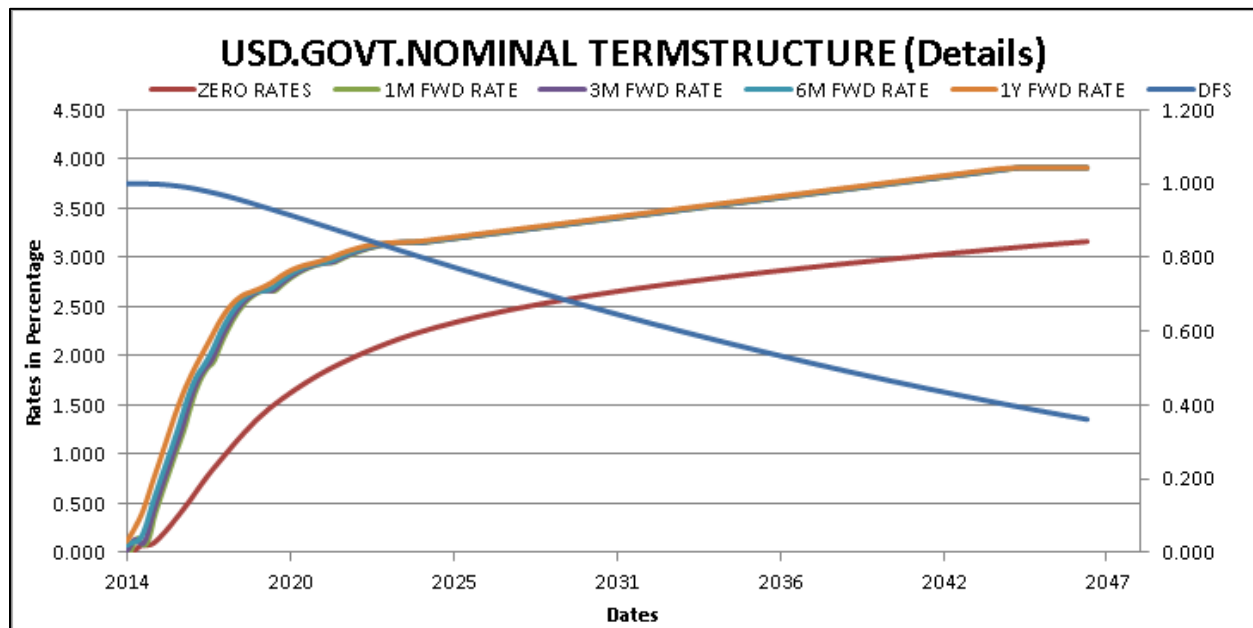
USD Treasury Curve Termstructure under LogLinear Interpolation.

This figure shows the treasury curve termstructure (discount factors, zero rates, 1M, 3M, 6M, 1Y forward rates) under loglinear interpolation.



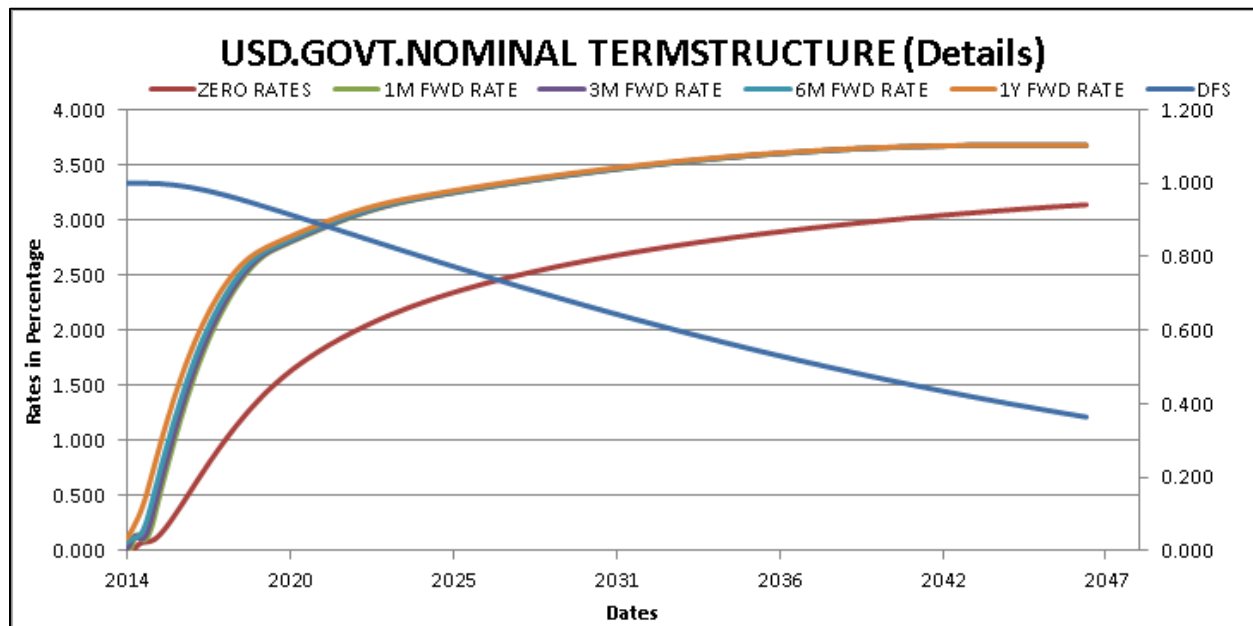
USD Treasury Curve Termstructure under LogParabolic Interpolation.

This figure shows the treasury curve termstructure (discount factors, zero rates, 1M, 3M, 6M, 1Y forward rates) under logparabolic interpolation.



USD Treasury Curve Termstructure under LogCubic Interpolation.

This figure shows the treasury curve termstructure (discount factors, zero rates, 1M, 3M, 6M, 1Y forward rates) under logcubic interpolation.



USD Treasury Curve Short End Termstructure under LogLinearCubic Interpolation.

This figure shows the treasury curve short end termstructure (discount factors, zero rates, 1M, 3M, 6M, 1Y forward rates) under loglinearcubic interpolation. The long end looks exactly same as logcubic interpolation.

